

Name:

Enrolment No:



UPES

End Semester Examination, May 2026

Course: Python Programming

Program: B.Tech. CSE

Course Code: CSEG1021

Semester: 2

Time : 03 hrs.

Max. Marks: 100

Instructions:

- Answer all the question
- No Calculators allowed

SECTION A
(5Qx4M=20Marks)

S. No.		Marks	CO
Q 1	Implement the below given flowchart. Provide the 2 test case and the sample output. <pre> graph TD START([START]) --> INPUT_SP[/INPUT SP/] INPUT_SP -.-> SP_note[SP represents the sale price] INPUT_SP --> PRINT_ENTER_N[/PRINT 'ENTER 'N' For National 'F' for Foreign'/] PRINT_ENTER_N --> INPUT_CHOICE[/INPUT CHOICE/] INPUT_CHOICE -.-> CHOICE_note[CHOICE holds either N or F] INPUT_CHOICE --> IS_CHOICE{IS CHOICE = 'N' ?} IS_CHOICE -- Yes --> ST_08[ST = SP * .08] IS_CHOICE -- No --> ST_18[ST = SP * .18] ST_08 --> NP[NP = SP + ST] ST_18 --> NP NP --> PRINT_NP[/PRINT NP/] PRINT_NP --> STOP([STOP]) </pre>	4	CO1
Q 2	An engineer needs a Python program that accepts a number from the user and calculates its power using the math module. Write the code for him.	4	CO2
Q 3	State the working of the assert statement with an example.	4	CO3
Q 4	“Do Python Support Constructor/Function Overloading?”- Justify the statement with suitable examples.	4	CO4
Q 5	Why are private data members important in object-oriented programming? Justify with an example.	4	CO4

SECTION B
(4Qx10M= 40 Marks)

Q 6	Classify the various datatypes of Python as mutable or immutable and justify your answer with examples.	10	CO1
Q 7	Write a Python program to find the square of numbers in a given list [2,5,8,11]. a) Using a lambda function with map(). b) Without using a lambda function.	10	CO2
Q 8	You are working with a dataset stored in a CSV file named company_ranks.csv using Pandas. The dataset contains information such as company_names, ranks, countries, revenues, and profits. Write a Pandas code to perform the following operations: a) Load the CSV file into a DataFrame and display its structure. b) Find the dimensions (shape) of the DataFrame. c) Handle missing data by: i. Identifying missing values in the dataset ii. Displaying the count of missing values in each column iii. Filling missing values in numerical columns (revenues, profit) with the mean iv. Filling missing values in categorical columns (country) with a default value like "Unknown" d) Select and display multiple columns (country and rank) e) Select rows where companyName is Apple or Microsoft, and display the country and rank. f) Create a filter to display companyNames with revenues > 100,000 and profit < 0.	10	CO5
Q 9	Instruction: Answer 9.a OR 9.b a) You are developing a login form using Tkinter with username and password fields. When the user clicks the login button, the system should validate the inputs and display a success or error message. Explain how event handling is used to process the button click. How will you bind the event and validate user input? OR b) You are developing a student record management system using Tkinter where student details are stored permanently in a database. The system should allow users to: i. Add new student records ii. View existing records iii. Update student details iv. Delete records Explain how Tkinter can be integrated with a database. How are events used to perform CRUD operations (Create, Read, Update, Delete)?	10	CO5

SECTION-C
(2Qx20M=40 Marks)

Q 10	Instruction: Answer 10.a,b,c OR 10.d,e a) Discuss the various mode while reading a file. (4 Marks) b) Write a Python program to create a file named "name.txt" and store a few names in it, with one name on each line. Based on the data in the file, perform the following operations: count the total number of names, count how many names start with a vowel, and find the longest name in the file. (6 Marks) c) Explain the concept of modules and packages in Python. Demonstrate how to create and import a module and organize multiple modules into a package with an example. (10 Marks) OR d) Discuss the various exception handling mechanisms in Python with suitable code. (10 Marks) e) Demonstrate with an example how file handling and exception handling work together in real-world applications. (10 Marks)	20	CO3
Q 11	a) A company wants to store employee information such as employee name, employee ID, and company name. The company name is common for all employees, whereas the employee name and employee ID are unique for each employee. (10 Marks) - Develop a Python program that defines a class using a class variable to store the company name and instance variables to store employee details. - Create objects for at least two employees and display their information - Create a factory method to count and display the number of employees enrolled. b) List and write programs to implement any four types of inheritance in python with suitable examples. (10 Marks)	20	CO4